# Simulation of Two-body Decay and Radioactive Equilibrium States

Numerical Methods and Programming Lab (PH5720)

Gandhar Vabale
EP22B047

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# Project Overview

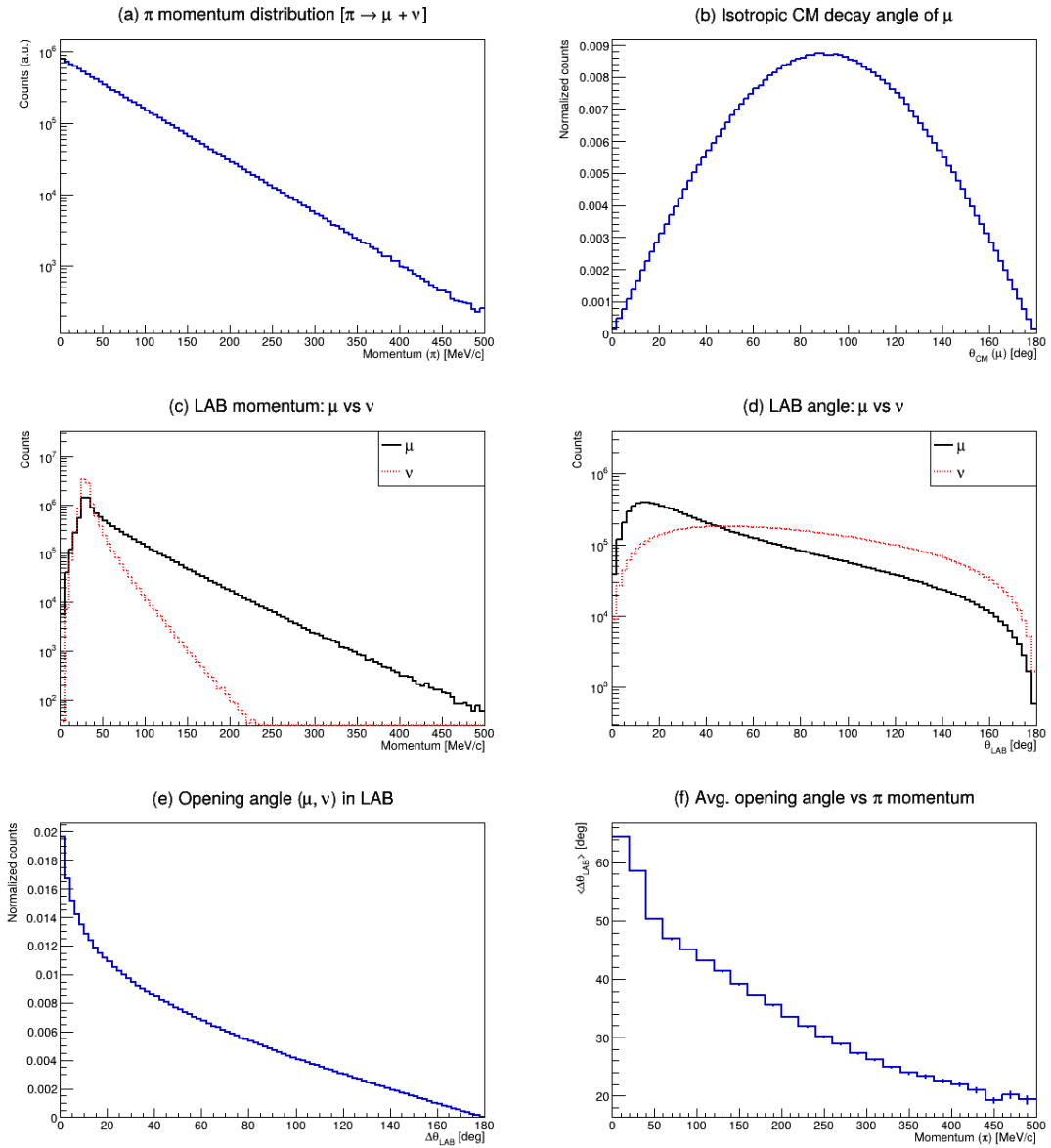
- Based on the paper: *"Simulation of two-body decay of a particle and radioactive equilibrium states"* .
- **Exercise I:** Simulation of $\pi \to \mu + \nu$ and $\pi \to e + \nu$ decay kinematics .
- **Exercise II:** Simulation of radioactive decay series $(P \to Q \to R)$ to study equilibrium states .
- **Numerical Framework:** Developed in C++ using the ROOT data analysis framework for Monte-Carlo generation and plotting .

# Exercise I: Implementation Details

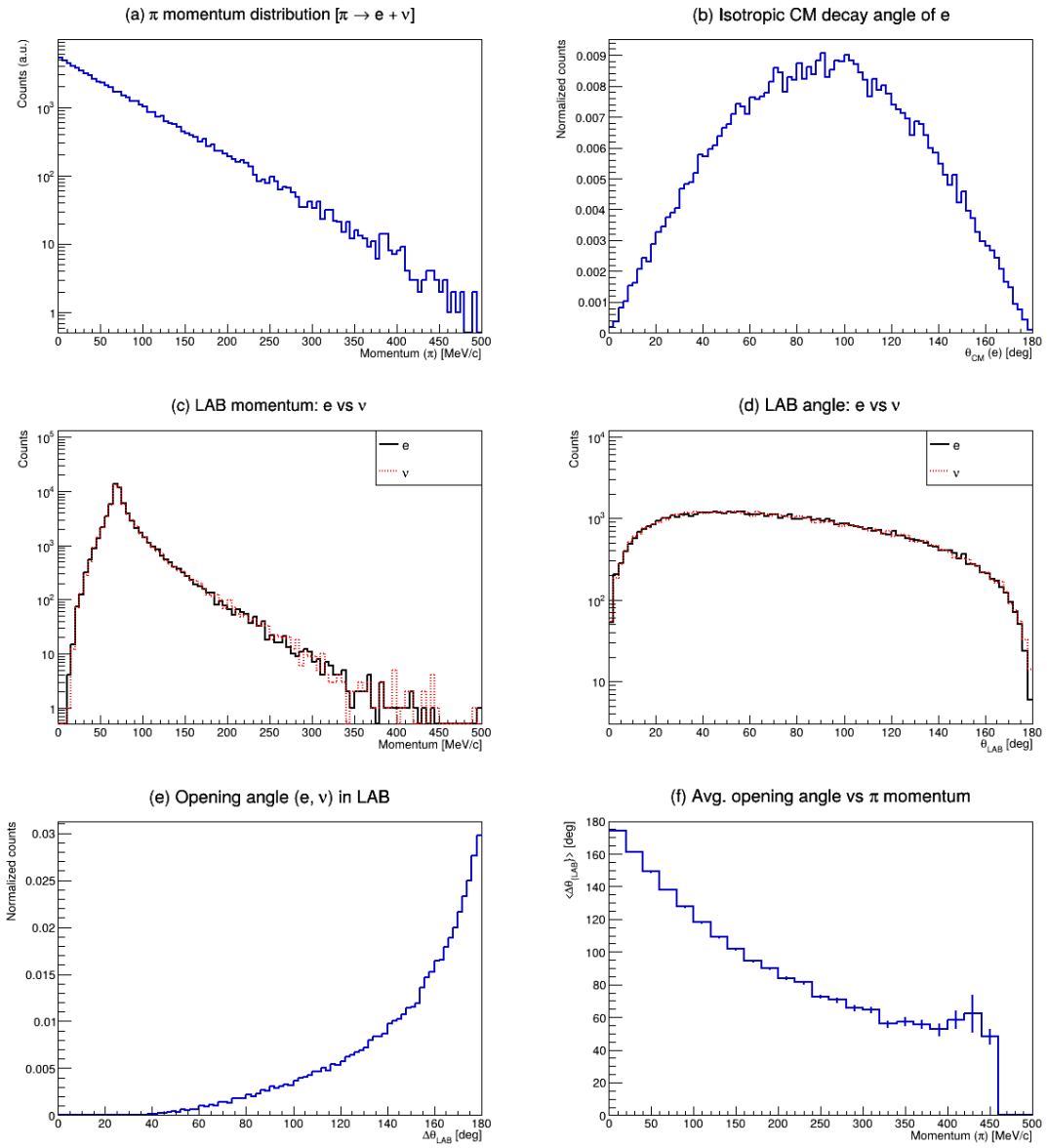
**Goal:** Simulate decay products in Center-of-Mass (CM) and Laboratory (LAB) frames.

- **Algorithm:**
  1. Generate $500M$ pions with an exponentially falling momentum spectrum .
  2. Assign masses based on Branching Ratios ($\pi \to \mu\nu \approx 99.98\%$) using a uniform random number .
  3. **CM Frame:** Isotropic decay simulated by generating $\cos\theta_{CM}$ uniformly between $[-1, 1]$ .
  4. **Lorentz Transformations:** Apply relativistic boosts to calculate momentum and angles in the LAB frame .

# Exercise I: Results ($\pi \to \mu + \nu$)



# Exercise I: Results ($\pi \to e + \nu$)



# Exercise II: Numerical Method

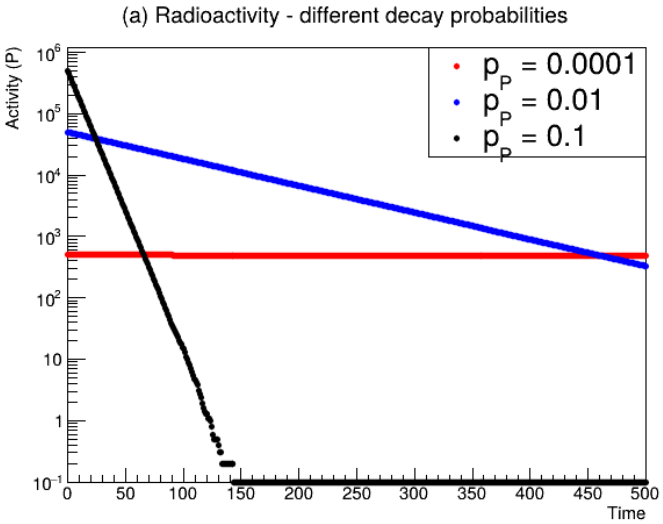
**Monte-Carlo Approach to Stochastic Decay:**

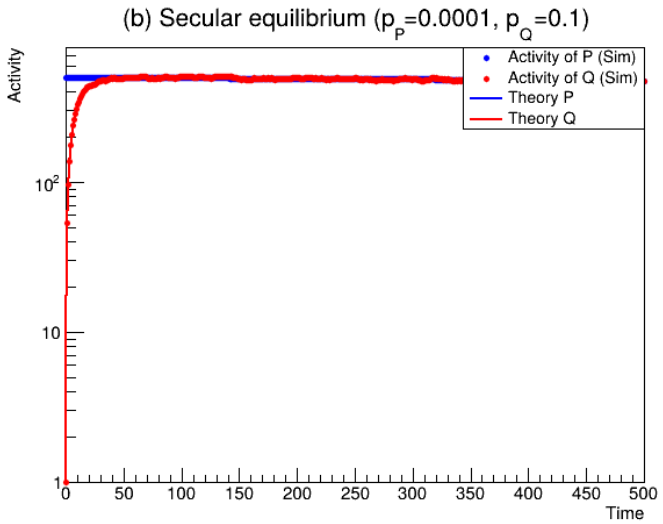
- Instead of solving ODEs directly, we simulate $N$ individual nuclei.
- For each nucleus, generate a uniform random number $r \in [0, 1]$ .
- If $r < p$ (decay probability), the nucleus decays .
- **Series Logic:** In each time step $\Delta t$, undecayed $P$ can become $Q$, and existing $Q$ can become $R$ (stable) based on probabilities $p_P$ and $p_Q$ .

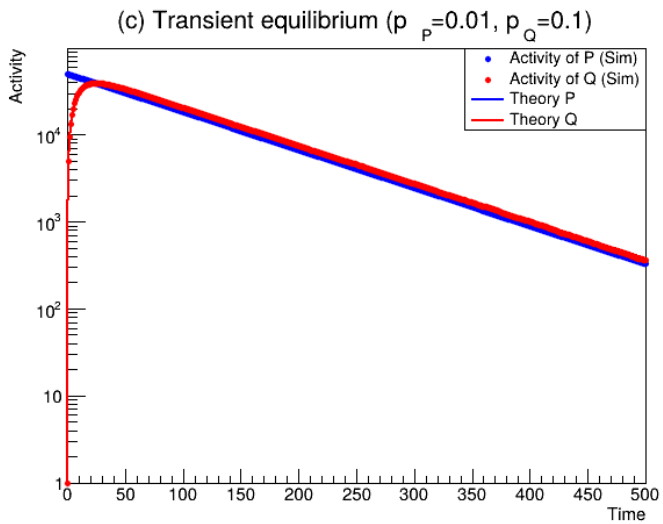
# Exercise II: Programming Specifics

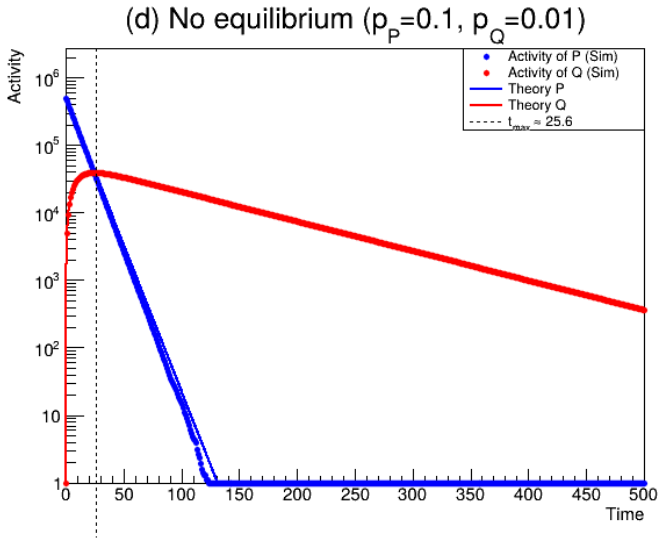
- **Parameters Used:**
  - Secular: $p_P = 0.0001, p_Q = 0.1$ .
  - Transient: $p_P = 0.01, p_Q = 0.1$ .
  - No Equilibrium: $p_P = 0.1, p_Q = 0.01$.
- **Visualization:** Simulation data is plotted using **point plots** (`TGraph` with option "P") to distinguish Monte-Carlo results from analytical theory lines .
- **Scaling:** Activities are tracked over 500 time units for $5 \times 10^6$ initial nuclei .

# Exercise II: Equilibrium Results



# Conclusion

- Successfully implemented two-body decay kinematics using Lorentz transformations .
- Verified that Monte-Carlo simulation of discrete nuclei accurately reproduces the macroscopic exponential decay laws and equilibrium conditions .

# Appendix: Reference Formulae (Exercise I)

**Center-of-Mass (CM) Frame Kinematics :**

- **Conservation of Momentum:** $\vec{p}_B + \vec{p}_C = 0 \Rightarrow \vec{p}_B = -\vec{p}_C$
- **Momentum Magnitude:**

$$|\vec{p}_{B,C}| = \frac{c}{2m_A}\sqrt{[m_A^2 - (m_B + m_C)^2] \times [m_A^2 - (m_B - m_C)^2]}$$

- **Isotropic Distribution:** $\cos\theta_{CM} \in [-1, 1]$
- **Momentum Components :**

$$p_B^{||} = p_B \cos\theta_{CM}, \quad p_B^{\perp} = p_B \sin\theta_{CM}$$

- **Energies :** $E_{B,C} = \sqrt{m_{B,C}^2 c^4 + |\vec{p}_{B,C}|^2 c^2}$

# Appendix: Reference Formulae (Exercise I)

**Laboratory (LAB) Frame Transformations :**

- **Lorentz Factors:** $\gamma = \frac{E_\pi}{m_\pi c^2}, \quad \beta = \sqrt{1 - 1/\gamma^2}$
- **Boosted Momentum (Parallel):** $p_B'^{||} = \gamma p_B \cos\theta_{CM} + \beta\gamma E_B$
- **Boosted Momentum (Perpendicular):** $p_B'^{\perp} = p_B \sin\theta_{CM}$
- **Boosted Energy:** $E_B' = \gamma E_B + \beta\gamma p_B \cos\theta_{CM}$
- **LAB Decay Angle:** $\theta_{LAB}^B = \cos^{-1}\left(\frac{p_B'^{||}}{|\vec{p}_B'|}\right)$

# Appendix: Reference Formulae (Exercise II)

**Radioactive Decay Series ($P \to Q \to R$) :**

- **Decay Law:** $N_P(t) = N_P^0 e^{-\lambda_P t}$
- **Rate Equation for Daughter (Q):** $\frac{dN_Q}{dt} = \lambda_P N_P - \lambda_Q N_Q$
- **Analytical Solution for $N_Q$:**

$$N_Q(t) = \frac{\lambda_P}{\lambda_Q - \lambda_P} N_P^0 (e^{-\lambda_P t} - e^{-\lambda_Q t})$$

- **Activity of Daughter ($M_Q$):**

$$M_Q(t) = \frac{\lambda_Q}{\lambda_Q - \lambda_P} M_P^0 (e^{-\lambda_P t} - e^{-\lambda_Q t})$$

- **Time of Maximum Activity ($t_{max}$) :**

$$t_{max} = \frac{\ln(\lambda_Q/\lambda_P)}{\lambda_Q - \lambda_P}$$

# Appendix: Reference Formulae (Exercise II)

**Equilibrium States :**

- **Secular Equilibrium ($\lambda_Q \gg \lambda_P, \lambda_P \to 0$):**

$$M_Q \approx M_P^0(1 - e^{-\lambda_Q t}) \quad \xrightarrow{t\to\infty} \quad M_Q = M_P$$

- **Transient Equilibrium ($\lambda_Q > \lambda_P$):**

$$M_Q = \frac{\lambda_Q}{\lambda_Q - \lambda_P} M_P^0 e^{-\lambda_P t}$$

- **No Equilibrium ($\lambda_Q < \lambda_P$):** Parent decays faster than daughter; daughter activity peaks at $t_{max}$ and then decays at its own rate .